# Donghyeon Soon

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## Education

**DGIST (Daegu Gyeongbuk Institute of Science & Technology)** — *Daegu, South Korea*
*B.S. in Computer & Electronics Engineering* — *Feb 2022 – Aug 2026 (expected)*

- Focus: Computer Science, Artificial Intelligence, 3D Vision, Robotics
- Coursework: Computer Vision, Machine Learning, Deep Learning, Graphics

## Research Interests

- 3D vision and generative modeling for reconstructing, animating, and generating dynamic 3D/4D content.
- Human and animal reconstruction using synthetic data pipelines, parametric models, and image/video-to-3D/4D methods.

## Experience

**Research Intern** — *Daegu, South Korea*
*Intelligent Systems and Learning Lab (Advisor: Prof. Daehee Park), DGIST* — *Dec 2025 – Present*

- Research on scene-aware human motion prediction, focusing on environment-aware future trajectory and pose estimation.

**Research Collaborator** — *Austin, Texas, USA (Remote)*
*The University of Texas at Austin (with Hezhen Hu)* — *Jun 2025 – Present*

- Collaborating on 3D/4D reconstruction, human and animal generative modeling, and expressive multi-person motion data.

**KIC Tech Frontier Program Participant** — *Berkeley, California, USA*
*UC Berkeley & KIC* — *Jun 2025 – Aug 2025*

- Conducted a startup project on real-time 3D human motion estimation and technology commercialization.

**Research Intern** — *Daegu, South Korea*
*Computer Vision Lab (Advisor: Prof. Sunghoon Im), DGIST* — *Jan 2025 – Feb 2025*

- Explored 3D Gaussian-based editing, focusing on Gaussian deformation and style-transfer techniques.
- Implemented a deformation-aware style-transfer pipeline for local geometry manipulation and appearance transfer in Gaussian Splatting radiance fields.

**Research Intern** — *Ulsan, South Korea*
*3D Vision & Robotics Lab (Advisor: Prof. Kyungdon Joo), UNIST* — *Jun 2023 – Dec 2024*

- Studied generative AI for animatable 3D reconstruction, Gaussian Splatting, and animal/human parametric modeling.
- Co-authored **DogRecon**, an animatable 3D Gaussian dog reconstruction method published in *IJCV 2025*.
- Contributed to pipeline implementation, module development, manuscript writing, and revision.

## Publications & Manuscripts

### Preprints / Under Review

**Feed-Forward Animatable Animal Reconstruction** — Under review

**Multimodal Clinical Prediction Research** — Under review

**Expressive Multi-Dancer Dataset** — Under review

## International

**Learning Scene-Grounded Interaction Priors for Scene-Aware Human Motion Prediction** 2026
**Donghyeon Soon***, Jiil Yoon*, Hyeondong Kim, Daehee Park†
*Advances in Neural Information Processing Systems (NeurIPS), 2026*
Project page

**AniGauss: Toward Animatable Animal Reconstruction from Single In-the-Wild Images via Topology-Aware Gaussians** 2026
Gyeongsu Cho, Changwoo Kang, **Donghyeon Soon**, Hezhen Hu, Kyungdon Joo†
*CV4Animals Workshop at CVPR 2026*
Paper / Project page

**WildAni4D: Towards 4D Animal Mesh Reconstruction** 2026
Gyeongsu Cho, Hezhen Hu, **Donghyeon Soon**, Changwoo Kang, Kyungdon Joo†
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026 Findings*

**Not Like Transformers: Drop the Beat Representation for Dance Generation with Mamba-Based Diffusion Model** 2026
Sangjune Park, Inhyeok Choi, **Donghyeon Soon**, Youngwoo Jeon, Kyungdon Joo†
*IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026*
*Also presented at ICCV 2025 Workshop on Generative Models for Audio-Visual Creation (Gen4AVC)*
Paper / Project page

**DogRecon: Canine Prior-Guided Animatable 3D Gaussian Dog Reconstruction From a Single Image** 2025
Gyeongsu Cho, Changwoo Kang, **Donghyeon Soon**, Kyungdon Joo†
*International Journal of Computer Vision (IJCV), 2025*
*Also presented at CVPR 2024 Workshop on Computer Vision for Animals (CV4Animals)*
Paper / Project page

## Domestic

**Consistent Image Generation via Aligned-Mean Flow Coupling** 2027
Minseok Oh, **Donghyeon Soon**, Jeong Hoon Choi, Jeong Tae Young
*Korea Artificial Intelligence Conference, 2027*

**Enhancing Foothold Planning for Locomotion on Curved Discrete Terrain** 2026
Taebok Lee*, **Donghyeon Soon***, Kyungdon Joo†
*ICROS, 2026 (Undergraduate Paper Award)*

**Improving Geometric Processing with Plane-aware Perturbation Uncertainty and Inference-time Control for Monocular Depth Estimation** 2026
**Donghyeon Soon***, Minseok Oh*, Sangchul Lee†
*Korea Artificial Intelligence Conference, 2026*

**CC-RAG: Confidence-Controlled Retrieval-Augmented Generation** 2026
**Donghyeon Soon***, Minseok Oh*, Sangchul Lee†
*Korea Artificial Intelligence Conference, 2026*

**Next-Scale Inversion Based Reference-Guided T2I Style Transfer** 2026
Minseok Oh*, **Donghyeon Soon***, Sangchul Lee†
*Korea Artificial Intelligence Conference, 2026 (Outstanding Paper Award)*

## Domestic (continued)

**Grid-based Local Homography Model for Panorama Stitching under Parallax** 2026
**Donghyeon Soon**, Minseok Oh, Taebok Lee
*Korea Artificial Intelligence Conference, 2026*

**Ghost Flare Localization and Removal Via Convex Sparse Template Selection** 2026
Minseok Oh*, **Donghyeon Soon***, Sangchul Lee†
*Korea Artificial Intelligence Conference, 2026*

**Enhancing Underwater Object Detection via Conditional Diffusion Model based Preprocessing and Augmentation** 2025
**Donghyeon Soon***, Taebok Lee*
*Korea Artificial Intelligence Conference, 2025 (Outstanding Paper Award)*
Paper

** Equal contribution. † Advisor/corresponding author.*

## Selected Projects

**Scene-aware Human Motion Prediction** *Dec 2025 – Present*
*DGIST ISL Lab*

- Developed a framework for human motion prediction that captures interactions with the surrounding environment.
- Targets improved future trajectory and pose estimation in scene-aware settings.

**Expressive Multi-Dancer Dataset** *2026*
*UT Austin & UNIST*

- Built a large-scale expressive multi-dancer dataset with automated annotations.
- Includes world-grounded multi-person SMPL-X motion, detailed hand articulation, facial expressions, audio, and structured text descriptions.

**AniGauss: Topology-Aware Animatable Animal Reconstruction** *2026*
*UNIST & UT Austin*

- Designed a feed-forward single-image animal reconstruction pipeline using canonical view generation and SMAL-topology-aware Gaussians.
- Project page: vision3d-lab.github.io/anigauss

**KIC Tech Frontier Program: Real-Time 3D Human Motion Estimation** *Jun 2025 – Aug 2025*
*UC Berkeley & KIC*

- Conducted a startup-oriented project on real-time 3D human motion estimation.
- Participated in entrepreneurship and technology commercialization activities through the KIC Tech Frontier Program.

## Skills

**Programming:** Python, C++, C, CUDA, Bash

**Frameworks & Tools:** PyTorch, OpenCV, Git, LaTeX, Gaussian Splatting pipelines

**Domains:** 3D/4D Vision, Generative Modeling, Diffusion Models, Gaussian Splatting, Human Motion Prediction, Human/Animal Reconstruction, Parametric Models (SMAL, SMPL, SMPL-X), Pose Estimation, Synthetic Data Generation, Robotics